

1.8 Integration

Mathematics — OCR A-Level (H240) — Paper 1: Pure Mathematics

Integration is the reverse process of differentiation. Where differentiation tells you how a quantity is changing, integration lets you recover the quantity from its rate of change — or, geometrically, compute the area under a curve. The two operations are bound together by the *fundamental theorem of calculus*, one of the most important results in mathematics. This chapter develops the techniques of indefinite and definite integration — power rule, standard integrals, substitution, integration by parts, partial fractions, and trigonometric identities — and uses them to find areas, volumes of revolution, and solutions to first-order separable differential equations. Every applied problem involving accumulation, total change, work, or volume ultimately rests on integration.

1. Integration as the Reverse of Differentiation

If differentiation takes a function $F(x)$ and produces its derivative $f(x) = F'(x)$, then integration runs the process backwards: given $f(x)$, find a function whose derivative is $f(x)$. Such a function is called an *antiderivative* of f . The essential subtlety is that antiderivatives are not unique — adding any constant to an antiderivative still yields an antiderivative, because the derivative of a constant is zero.

Indefinite integral (antiderivative). An *antiderivative* of $f(x)$ is any function $F(x)$ with $F'(x) = f(x)$. The set of all antiderivatives is denoted $\int f(x) dx = F(x) + C$, where C is an arbitrary constant — the *constant of integration*. The symbol $\int$ is an elongated "S" (from "sum"); dx indicates the variable of integration.

Why the constant? If $F'(x) = f(x)$ and $G(x) = F(x) + C$ for any constant C , then $G'(x) = F'(x) + 0 = f(x)$ as well. So F and G are equally valid antiderivatives. Conversely, any two antiderivatives of f differ by a constant (this is a consequence of the mean value theorem). The "+C" captures this ambiguity.

Worked Example 1 — Why the constant matters

1. Consider $f(x) = 2x$.
2. $F(x) = x^2$ has derivative $2x$ ✓
3. $G(x) = x^2 + 7$ has derivative $2x$ ✓
4. $H(x) = x^2 - 3.4$ has derivative $2x$ ✓
5. All three are antiderivatives of $2x$ — they differ only by a constant.
6. We capture all of them at once by writing:
7. $\int 2x \, dx = x^2 + C$.

// The $+C$ is mandatory in every indefinite integral. Losing it loses a mark.

Exam tip — always write $+C$ for indefinite integrals. OCR mark schemes list the constant of integration as a standalone awarded mark on almost every indefinite-integral question. Forgetting it is the single most common error at A-level. The only place you don't need $+C$ is in a *definite* integral, because the constant cancels when you subtract $F(a)$ from $F(b)$.

2. The Power Rule for Integration

Reversing the power rule $\frac{d}{dx}(x^{n+1}) = (n+1)x^n$ gives the integration power rule. Divide by $n+1$ to isolate x^n :

Power rule for integration. For any real $n \neq -1$, $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$. In words: *add one to the power, divide by the new power*. The exclusion $n = -1$ is needed because division by zero would occur. The case $n = -1$ is handled by the logarithm: $\int \frac{1}{x} \, dx = \ln|x| + C$. The absolute value is essential — $\ln$ is only defined for positive arguments, but $1/x$ is defined for all $x \neq 0$.

The power rule extends term-by-term via linearity: for constants a, b and integrable functions f, g ,

$$\int [af(x) + bg(x)] \, dx = a \int f(x) \, dx + b \int g(x) \, dx.$$

Worked Example 2 — Polynomial with mixed powers

Find $\int (3x^2 - 4x + 1) dx$.

1. Integrate term by term using the power rule:
2. $\int 3x^2 dx = 3 \cdot (x^3/3) = x^3$.
3. $\int -4x dx = -4 \cdot (x^2/2) = -2x^2$.
4. $\int 1 dx = x$ (since $1 = x^0$, add 1 to get $x^1/1$.)
5. Combine and add the constant:
6. $\int (3x^2 - 4x + 1) dx = x^3 - 2x^2 + x + C$.
7. Check by differentiating: $d/dx (x^3 - 2x^2 + x + C) = 3x^2 - 4x + 1$ ✓.

// Always verify by differentiating — it catches arithmetic errors in seconds.

3. Standard Integrals

Reversing the standard derivatives gives a short list of integrals that must be memorised. Each is just a differentiation rule read backwards.

Function $f(x)$	Integral $\int f(x) dx$	Notes
$x^n (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$	Any real n except -1 .
$1/x$	$\ln x + C$	Absolute value for $x < 0$.
$\sin x$	$-\cos x + C$	Minus sign — easily missed.
$\cos x$	$\sin x + C$	No minus sign.
$\sec^2 x$	$\tan x + C$	Reverses $(\tan x)' = \sec^2 x$.
e^x	$e^x + C$	Unique: same as function.
e^{kx}	$\frac{1}{k}e^{kx} + C$	Divide by the inner coefficient.

$\sin(kx)$	$-\frac{1}{k} \cos(kx) + C$	Chain-rule reversal.
$\cos(kx)$	$\frac{1}{k} \sin(kx) + C$	Chain-rule reversal.
$\frac{1}{ax+b}$	$\frac{1}{a} \ln ax+b + C$	Linear inner — divide by a .

Exam tip — watch for the sign on sin and cos. It is easy to mix up which of sin and cos picks up a minus sign on integration. The safest rule: *differentiate your answer to check*. For $\int \sin x \, dx = -\cos x + C$: differentiate $-\cos x$ to get $-(-\sin x) = \sin x \checkmark$. For $\int \cos x \, dx = \sin x + C$: differentiate $\sin x$ to get $\cos x \checkmark$. A five-second check prevents a lost mark.

4. Definite Integrals and the Fundamental Theorem

An indefinite integral produces a family of antiderivatives. A *definite* integral produces a number — the net change of an antiderivative between two specified limits, equivalently the (signed) area between the curve and the x-axis.

Definite integral. If F is an antiderivative of f on $[a, b]$, then the definite integral of f from a to b is $\int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a)$. The numbers a and b are the lower and upper *limits of integration*. No constant of integration is written — it would cancel in $F(b) + C - (F(a) + C)$.

Fundamental theorem of calculus. If f is continuous on $[a, b]$, then $F(x) = \int_a^x f(t) \, dt$ is differentiable on (a, b) and $F'(x) = f(x)$. Consequently, if F is any antiderivative of f on $[a, b]$, then $\int_a^b f(x) \, dx = F(b) - F(a)$. This theorem ties the two branches of calculus together: the integral is the inverse of the derivative.

Useful properties of definite integrals:

- **Swap limits, change sign:** $\int_b^a f(x) \, dx = -\int_a^b f(x) \, dx$.
- **Adjacent interval additivity:** $\int_a^b f \, dx + \int_b^c f \, dx = \int_a^c f \, dx$.
- **Zero interval:** $\int_a^a f(x) \, dx = 0$.

- **Linearity:** $\int_a^b [\alpha f + \beta g] dx = \alpha \int_a^b f dx + \beta \int_a^b g dx$.

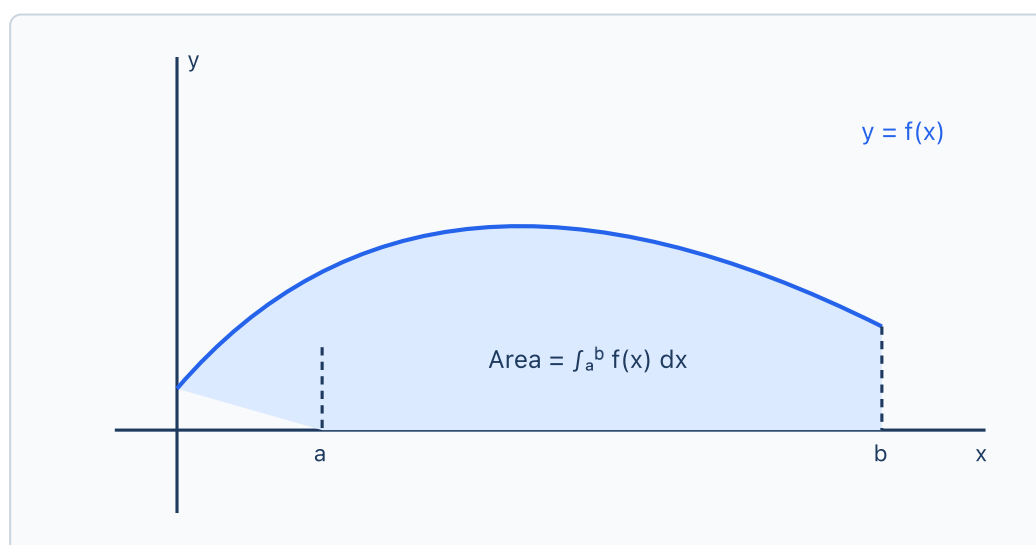
Worked Example 3 — Evaluate $\int_1^3 (2x + 1) dx$

1. Find an antiderivative:
2. $\int (2x + 1) dx = x^2 + x$ (+C, but we may drop C for definite integrals)
3. Apply the limits:
4. $[x^2 + x]_1^3 = (3^2 + 3) - (1^2 + 1)$
5. $= (9 + 3) - (1 + 1)$
6. $= 12 - 2$
7. $= 10$.
8. $\int_1^3 (2x + 1) dx = 10$.

// Geometric check: this integral is the area of a trapezium under $y = 2x + 1$
 // from $x = 1$ to $x = 3$ — heights 3 and 7, width 2, area = $\frac{1}{2}(3+7)(2) = 10$ ✓.

5. Area Under a Curve

If $f(x) \geq 0$ throughout $[a, b]$, the definite integral $\int_a^b f(x) dx$ equals the area enclosed between the curve $y = f(x)$, the x-axis, and the vertical lines $x = a$ and $x = b$. If $f(x) < 0$ somewhere on the interval, the integral measures a *signed* area — portions below the axis contribute negative amounts.



Exam tip — "area" versus "integral" can differ. If the curve dips below the x-axis on the interval of interest, the integral gives signed area (parts below count as negative) while the *geometric area* is always positive. For geometric area you must (i) find where f crosses the axis inside $[a, b]$, (ii) split the interval there, and (iii) take the absolute value of each piece before summing. Always sketch the curve first.

Worked Example 4 — Signed vs unsigned area

Compute $\int_0^3 (x^2 - 2x) dx$ and also find the total geometric area between $y = x^2 - 2x$ and the x-axis on $[0, 3]$.

1. Antiderivative: $\int (x^2 - 2x) dx = x^3/3 - x^2$.
 2. Signed integral:
 3. $[x^3/3 - x^2]_0^3 = (9 - 9) - (0 - 0) = 0$.
 4. The signed integral is ZERO — the positive and negative parts cancel.
 - 5.
 6. Find the zero of the integrand inside $[0, 3]$:
 7. $x^2 - 2x = x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$.
 8. On $(0, 2)$ the curve is BELOW the axis ($f(1) = -1 < 0$).
 9. On $(2, 3)$ the curve is ABOVE the axis ($f(2.5) = 1.25 > 0$).
 - 10.
 11. Split and take absolute values:
 12. $\int_0^2 (x^2 - 2x) dx = (8/3 - 4) - 0 = -4/3 \Rightarrow |-4/3| = 4/3$.
 13. $\int_2^3 (x^2 - 2x) dx = (9 - 9) - (8/3 - 4) = 0 - (-4/3) = 4/3$.
 14. Total geometric area = $4/3 + 4/3 = 8/3$.
- // The integral is 0 but the area is $8/3$ — two different numbers from the same

6. Area Between Curves

To find the area enclosed between two curves $y = f(x)$ (top) and $y = g(x)$ (bottom) on $[a, b]$, integrate the difference:

$$\text{Area} = \int_a^b [f(x) - g(x)] dx, \quad \text{where } f \geq g \text{ on } [a, b].$$

The limits a, b are usually the x -coordinates of the intersection points, found by solving $f(x) = g(x)$. If the roles of "top" and "bottom" swap within the interval, split the integral at the crossover.

Worked Example 5 — Area enclosed by $y = x^2$ and $y = 4$

1. Find intersection points: $x^2 = 4 \Rightarrow x = \pm 2$.
2. On $(-2, 2)$ the line $y = 4$ is ABOVE the parabola $y = x^2$.
3. (top – bottom) = $4 - x^2$.
4. Area = $\int_{-2}^2 (4 - x^2) dx$
5. = $[4x - x^3/3]_{-2}^2$
6. = $(8 - 8/3) - (-8 + 8/3)$
7. = $(24/3 - 8/3) - (-24/3 + 8/3)$
8. = $16/3 - (-16/3)$
9. = $32/3$.
10. Area enclosed = $32/3$ square units.

// By symmetry of the region in the y -axis, $2 \cdot \int_0^2 (4 - x^2) dx = 2 \cdot 16/3 = 32/3$

7. Integration by Substitution

Substitution is the integration analogue of the chain rule. If a composite function $f(g(x))$ has been multiplied by the derivative $g'(x)$ of its inner function, the substitution $u = g(x)$ collapses the integrand into a pure function of u .

Integration by substitution. If $u = g(x)$ is differentiable with $du = g'(x) dx$, then $\int f(g(x)) g'(x) dx = \int f(u) du$. For a definite integral, the limits must also be transformed: when $x = a$, set $u = g(a)$; when $x = b$, set $u = g(b)$. Equivalently, you may substitute back to x before applying the original limits.

Recognising when substitution helps: the integrand should contain a function of $g(x)$ alongside (a constant multiple of) the derivative $g'(x)$. If you spot $g'(x)$ multiplying $f(g(x))$, substitute $u = g(x)$.

Worked Example 6 — Substitution with $u = x^2 + 1$

Find $\int \frac{2x}{x^2 + 1} dx$.

1. Let $u = x^2 + 1$. Then $du/dx = 2x$, so $du = 2x dx$.
2. The integrand contains $2x dx$ in the numerator – perfect.
3. Rewrite:
4. $\int (2x / (x^2 + 1)) dx = \int (1/u) du$.
5. Integrate:
6. $\int (1/u) du = \ln|u| + C$.
7. Substitute back:
8. $= \ln|x^2 + 1| + C = \ln(x^2 + 1) + C$ (since $x^2 + 1 > 0$ always).

// Template: $\int f'(x) / f(x) dx = \ln|f(x)| + C$ – a very common pattern.

Worked Example 7 — Substitution with $u = x^2$

Find $\int 2x e^{x^2} dx$.

1. Let $u = x^2$. Then $du = 2x dx$.
2. Rewrite:
3. $\int 2x e^{(x^2)} dx = \int e^u du$.
4. Integrate:
5. $\int e^u du = e^u + C$.
6. Back-substitute:
7. $= e^{(x^2)} + C$.
8. Check by differentiation:
9. $d/dx [e^{(x^2)}] = e^{(x^2)} \cdot 2x \checkmark$.

Worked Example 8 — Linear substitution

Find $\int \cos(3x + 1) dx$.

1. Let $u = 3x + 1$. Then $du = 3 dx$, so $dx = du/3$.
2. Rewrite:
3. $\int \cos(3x + 1) dx = \int \cos u \cdot (1/3) du = (1/3) \int \cos u du$.
4. Integrate:
5. $(1/3) \sin u + C$.
6. Back-substitute:
7. $= (1/3) \sin(3x + 1) + C$.

// For any LINEAR inner function $ax + b$, the substitution just adds a factor of

Exam tip — for definite integrals, change the limits too. When you substitute $u = g(x)$ in $\int_a^b f(g(x))g'(x) dx$, the limits a, b are values of x . Under the new variable they become $g(a), g(b)$. Either (i) change the limits and never return to x , or (ii) keep the original limits and substitute back to x before evaluating. Mixing the two — evaluating u -limits on an x -expression — is a frequent and costly error.

8. Integration by Parts

Parts is the integration analogue of the product rule. Starting from $(uv)' = u'v + uv'$ and integrating both sides:

Integration by parts. If u, v are differentiable functions of x , then $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$, often written compactly as $\int u dv = uv - \int v du$. The technique trades one integral for another; for it to help, the new integral $\int v du$ must be simpler than the original.

Choosing u and dv : you want u to simplify on differentiation, and dv to remain integrable. A standard mnemonic is **LIATE**:

Logarithmic $\prec$ Inverse trig $\prec$ Algebraic $\prec$ Trig $\prec$ Exponential

Whichever type appears earlier in this list, set it as u . The remaining factor becomes dv .

Integrand type	Let $u =$	Let $dv =$	Reason
$x^n \ln x$	$\ln x$ (L)	$x^n dx$	$\ln x$ simplifies on differentiation.
$x^n e^{kx}$	x^n (A)	$e^{kx} dx$	x^n becomes a polynomial of lower degree.
$x^n \sin x$ or $x^n \cos x$	x^n (A)	$\text{trig } dx$	Polynomial degree decreases.
$\ln x$ (alone)	$\ln x$ (L)	dx	Differentiating $\ln x$ gives $1/x$.

Worked Example 9 — $\int x e^x dx$

1. LIATE: algebraic x before exponential e^x , so let $u = x$, $dv = e^x dx$.
2. Then $du = dx$ and $v = \int e^x dx = e^x$.
3. Apply parts:
4. $\int x e^x dx = uv - \int v du$
5. $= x e^x - \int e^x dx$
6. $= x e^x - e^x + C$
7. $= e^x (x - 1) + C$.
8. Check: $d/dx [e^x (x - 1)] = e^x (x - 1) + e^x = x e^x \checkmark$.

Worked Example 10 — $\int x \cos x dx$

1. LIATE: algebraic before trig. $u = x$, $dv = \cos x dx$.
2. $du = dx$; $v = \sin x$.
3. Parts:
4. $\int x \cos x dx = x \sin x - \int \sin x dx$
5. $= x \sin x - (-\cos x) + C$

6. $\quad = x \sin x + \cos x + C.$
7. Check: $d/dx [x \sin x + \cos x] = \sin x + x \cos x - \sin x = x \cos x \checkmark.$

Worked Example 11 — $\int \ln x \, dx$ (the "trick")

1. No obvious product — but write $\ln x = (\ln x) \cdot 1.$
2. Let $u = \ln x$, $dv = 1 \, dx.$
3. $du = (1/x) \, dx$; $v = x.$
4. Parts:
5. $\int \ln x \, dx = x \ln x - \int x \cdot (1/x) \, dx$
6. $\quad = x \ln x - \int 1 \, dx$
7. $\quad = x \ln x - x + C$
8. $\quad = x (\ln x - 1) + C.$
9. Check: $d/dx [x \ln x - x] = \ln x + 1 - 1 = \ln x \checkmark.$

// The "dv = dx" trick works for any single logarithm or inverse trig.

Worked Example 12 — $\int x^2 e^x \, dx$ (parts twice)

1. First pass. $u = x^2$, $dv = e^x \, dx \Rightarrow du = 2x \, dx$, $v = e^x.$
2. $\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx$
3. $\quad = x^2 e^x - 2 \int x e^x \, dx.$
4. Use Worked Example 9: $\int x e^x \, dx = e^x (x - 1) + C_1.$
5. Substitute:
6. $\int x^2 e^x \, dx = x^2 e^x - 2 [e^x (x - 1)] + C$
7. $\quad = x^2 e^x - 2x e^x + 2 e^x + C$
8. $\quad = e^x (x^2 - 2x + 2) + C.$
9. Check: $d/dx [e^x (x^2 - 2x + 2)] = e^x (x^2 - 2x + 2) + e^x (2x - 2)$
10. $\quad = e^x [x^2 - 2x + 2 + 2x - 2] = x^2 e^x \checkmark.$

// Rule of thumb: $\int x^n e^{(kx)} \, dx$ needs parts applied n times.

Exam tip — LIATE is a guideline, not a law. It works in the vast majority of A-level integrals, but if LIATE gives an integrand that does not simplify, try the other way. The real test is whether $\int v \, du$ is simpler than $\int u \, dv$. Also, for integrands like $\int x \ln x \, dx$, LIATE sends you to $u = \ln x$ (correct), but always double-check by trying to integrate the alternative to see if it terminates.

9. Integration Using Partial Fractions

A rational function $P(x)/Q(x)$ (with $\deg P < \deg Q$) can often be decomposed into a sum of simpler fractions. Each piece integrates to a logarithm or power. For distinct linear factors in the denominator:

$$\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}, \quad \int \frac{1}{x-a} \, dx = \ln|x-a| + C.$$

Worked Example 13 — Partial fractions

Find $\int \frac{3x+5}{(x-1)(x+2)} \, dx$.

1. Write: $(3x+5) / [(x-1)(x+2)] = A/(x-1) + B/(x+2)$.
2. Multiply through:
3. $3x+5 = A(x+2) + B(x-1)$.
4. Cover-up at $x=1$: $3(1)+5 = A(3) \Rightarrow 8 = 3A \Rightarrow A = 8/3$.
5. Cover-up at $x=-2$: $3(-2)+5 = B(-3) \Rightarrow -1 = -3B \Rightarrow B = 1/3$.
6. Rewrite the integrand:
7. $(3x+5) / [(x-1)(x+2)] = (8/3)/(x-1) + (1/3)/(x+2)$.
8. Integrate term by term:
9. $\int \dots \, dx = (8/3) \ln|x-1| + (1/3) \ln|x+2| + C$.

// Always write the cover-up values at the "bad" points of the denominator.
 // Combine the two logs if asked: $= (1/3) \ln|(x-1)^8 (x+2)| + C$.

10. Integration Using Trigonometric Identities

Many trigonometric integrands don't match the standard forms directly, but they can be rewritten using identities. The most useful identities for integration are:

Identity	Useful for
$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$	$\int \cos^2 x \, dx$
$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$	$\int \sin^2 x \, dx$
$\sin 2x = 2 \sin x \cos x$	$\int \sin x \cos x \, dx$
$\sec^2 x = 1 + \tan^2 x$	$\int \tan^2 x \, dx$
Product-to-sum, e.g. $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$	$\int \sin(mx) \cos(nx) \, dx$

Worked Example 14 — $\int \cos^2 x \, dx$

1. Apply the double-angle identity $\cos^2 x = (1 + \cos 2x)/2$.
 2. Rewrite:
 3. $\int \cos^2 x \, dx = \int \frac{1}{2}(1 + \cos 2x) \, dx$
 4. $= \frac{1}{2} \int 1 \, dx + \frac{1}{2} \int \cos 2x \, dx$
 5. $= \frac{1}{2} x + \frac{1}{2} \cdot \frac{1}{2} \sin 2x + C$
 6. $= x/2 + (\sin 2x)/4 + C.$
 7. Check: $d/dx [x/2 + (\sin 2x)/4] = 1/2 + (\cos 2x)/2 = (1 + \cos 2x)/2 = \cos^2 x$
- // The same trick with $\sin^2 x = (1 - \cos 2x)/2$ gives $\int \sin^2 x \, dx = x/2 - (\sin 2x)/4 + C$

Worked Example 15 — Product-to-sum

Find $\int \sin(2x) \cos(3x) \, dx$.

1. Apply product-to-sum:
2. $\sin A \cos B = \frac{1}{2} [\sin(A + B) + \sin(A - B)].$
3. $\sin(2x) \cos(3x) = \frac{1}{2} [\sin(5x) + \sin(-x)]$

$$\begin{aligned}
 4. & \quad = \frac{1}{2} \sin(5x) - \frac{1}{2} \sin x. \quad (\sin(-x) = -\sin x) \\
 5. & \quad \text{Integrate:} \\
 6. & \quad \int \sin(2x) \cos(3x) dx = \int \left[\frac{1}{2} \sin(5x) - \frac{1}{2} \sin x \right] dx \\
 7. & \quad = \frac{1}{2} \cdot \left(-\frac{1}{5}\right) \cos(5x) - \frac{1}{2} \cdot (-\cos x) + C \\
 8. & \quad = -(1/10) \cos(5x) + \frac{1}{2} \cos x + C.
 \end{aligned}$$

Exam tip — $\cos^2 x$ and $\sin^2 x$ cannot be integrated directly. Students often try $\int \cos^2 x dx = \frac{\cos^3 x}{3}$ (applying the power rule as if $\cos x$ were x) — this is wrong, because differentiating $\cos^3 x/3$ gives $-\cos^2 x \sin x$, not $\cos^2 x$. The only A-level-safe method is to linearise with the double-angle identity first.

11. Volumes of Revolution

Scope note: volumes of revolution are not in the OCR A-Level Mathematics (H240) specification. They are Further Mathematics (H245) content and are included here for enrichment only.

Rotating the region between a curve $y = f(x)$, the x-axis, and the lines $x = a$, $x = b$ through a full turn about the x-axis sweeps out a solid whose volume can be computed by integration. The thin-disk approximation: each thin strip of width dx at position x becomes a disk of radius $y = f(x)$ and thickness dx , contributing volume $\pi y^2 dx$. Integrating gives:

Solid of revolution — volume formula. The solid obtained by rotating the curve $y = f(x)$ around the x-axis from $x = a$ to $x = b$ has volume $V = \pi \int_a^b y^2 dx = \pi \int_a^b [f(x)]^2 dx$. For rotation about the y-axis of the curve $x = g(y)$ from $y = c$ to $y = d$, $V = \pi \int_c^d x^2 dy = \pi \int_c^d [g(y)]^2 dy$.

Axis of revolution	Variable of integration	Volume formula
x-axis	x	$V = \pi \int_a^b y^2 dx$
y-axis	y	$V = \pi \int_c^d x^2 dy$

Worked Example 16 — Volume of revolution of $y = \sin x$ about the x-axis, $0 \leq x \leq \pi$

1. Apply the disk formula:
2. $V = \pi \int_0^\pi y^2 \, dx = \pi \int_0^\pi \sin^2 x \, dx.$
3. Use $\sin^2 x = (1 - \cos 2x)/2$:
4. $V = \pi \int_0^\pi \frac{1}{2}(1 - \cos 2x) \, dx$
5. $= (\pi/2) [x - (\sin 2x)/2]_0^\pi.$
6. Evaluate:
7. at $x = \pi$: $\pi - (\sin 2\pi)/2 = \pi - 0 = \pi.$
8. at $x = 0$: $0 - (\sin 0)/2 = 0.$
9. $V = (\pi/2) \cdot \pi = \pi^2/2.$

// Volume of the "lemon" obtained by rotating one arch of a sine about the x-axis
 // Numerical: $\pi^2/2 \approx 4.9348$ cubic units.

12. First-Order Separable Differential Equations

A differential equation relates a function to its derivative(s). The simplest useful case is a *first-order separable* equation in which the derivative factors into a function of x times a function of y :

First-order separable differential equation. An equation of the form $\frac{dy}{dx} = f(x)g(y)$ is called *separable*. Separate the variables by treating dy/dx as a quotient of differentials: $\frac{1}{g(y)} \, dy = f(x) \, dx \quad \Longrightarrow \quad \int \frac{1}{g(y)} \, dy = \int f(x) \, dx + C.$ The resulting equation (after integrating both sides) is called the *general solution*. An *initial condition* $y(x_0) = y_0$ pins C down to give the *particular solution*.

Worked Example 17 — Solve $dy/dx = 2y$ with $y(0) = 3$

1. Separate variables (valid while $y \neq 0$):
2. $(1/y) \, dy = 2 \, dx.$
3. Integrate both sides:

4. $\int (1/y) dy = \int 2 dx$
5. $\ln|y| = 2x + C.$
6. Exponentiate:
7. $|y| = e^{(2x + C)} = e^C \cdot e^{(2x)}.$
8. $y = A \cdot e^{(2x)}$ ($A = \pm e^C$, any nonzero constant.)
9. Apply the initial condition $y(0) = 3$:
10. $3 = A \cdot e^0 = A \Rightarrow A = 3.$
11. Particular solution: $y = 3 e^{(2x)}.$
12. Check: $dy/dx = 6 e^{(2x)} = 2 \cdot 3 e^{(2x)} = 2y \checkmark. y(0) = 3 \checkmark.$

// dy/dx proportional to y gives exponential growth – the classic population /
 // radioactive-decay model.

Worked Example 18 — Solve $dy/dx = xy$ with $y(0) = 1$

1. Separate:
2. $(1/y) dy = x dx.$
3. Integrate:
4. $\int (1/y) dy = \int x dx$
5. $\ln|y| = x^2/2 + C.$
6. Exponentiate:
7. $y = A \cdot e^{(x^2/2)}.$
8. Initial condition $y(0) = 1$:
9. $1 = A \cdot e^0 \Rightarrow A = 1.$
10. Particular solution: $y = e^{(x^2/2)}.$
11. Check: $dy/dx = e^{(x^2/2)} \cdot x = x y \checkmark.$

13. Applications in Mechanics (SUVAT Links)

Integration provides the link between acceleration, velocity, and displacement for a particle in motion along a line. If $a(t)$ is the acceleration as a function of time:

$$v(t) = \int a(t) dt + C_1, \quad s(t) = \int v(t) dt + C_2.$$

The constants C_1, C_2 are fixed by initial conditions (usually $v(0) = u$ the initial velocity, and $s(0) = 0$ the starting position). For constant acceleration a , integrating gives the familiar SUVAT equations:

$$v = u + at, \quad s = ut + \frac{1}{2}at^2.$$

The power of integration is that it handles variable acceleration too. If $a(t) = 6 - 2t$ and a particle starts from rest, then $v(t) = 6t - t^2$ and $s(t) = 3t^2 - t^3/3$ — no SUVAT formula applies, but integration still gives the answer immediately.

14. Applied Worked Examples

Worked Example 19 — $\int x \ln x \, dx$ (parts with LIATE)

1. LIATE: logarithmic before algebraic, so $u = \ln x$, $dv = x \, dx$.
2. $du = (1/x) \, dx$; $v = x^2/2$.
3. Parts:
4. $\int x \ln x \, dx = (x^2/2) \ln x - \int (x^2/2)(1/x) \, dx$
5. $= (x^2/2) \ln x - \int (x/2) \, dx$
6. $= (x^2/2) \ln x - x^2/4 + C$
7. $= (x^2/4)(2 \ln x - 1) + C$.
8. Check: $d/dx [(x^2/2) \ln x - x^2/4] = x \ln x + x^2/2 \cdot (1/x) - x/2$
9. $= x \ln x + x/2 - x/2 = x \ln x \checkmark$.

15. Summary — When to Use Which Technique

Integrand pattern	Technique	Key formula
x^n or polynomial	Power rule	$\int x^n \, dx = x^{n+1}/(n+1) + C$
Standard trig / exp / log	Quote standard integral	See table in §3.

$f(g(x)) \cdot g'(x)$	Substitution $u = g(x)$	$\int f(u) du$
Product of different function types (algebraic $\times$ exp / trig / log)	Integration by parts, with LIATE	$\int u dv = uv - \int v du$
Rational function $P(x)/Q(x)$ ($\deg P < \deg Q$)	Partial fractions	Decompose, integrate each piece to $\ln$.
$\sin^2, \cos^2, \sin mx \cos nx$	Trig identity (linearise)	Double-angle or product-to-sum.
Area between f and g	Definite integral of difference	$\int_a^b (f - g) dx$
Volume of revolution about x-axis	Disk method	$V = \pi \int_a^b y^2 dx$
$dy/dx = f(x)g(y)$	Separate and integrate	$\int dy/g(y) = \int f(x) dx + C$

Chapter-wide checklist for Integration.

- **Always write $+C$** on every indefinite integral.
- **For definite integrals, no C — but always show $[F(x)]_a^b$** and the subtraction $F(b) - F(a)$.
- **Substitution: change the limits** if it's a definite integral, or substitute back if indefinite.
- **Parts: use LIATE** to pick u . Check whether $\int v du$ is actually simpler.
- **Sign of area:** if the curve dips below the axis, split the integral at the zeros.
- **For area between curves:** find intersection points first, then (top – bottom).
- **$\cos^2 x$ and $\sin^2 x$:** always linearise with the double-angle formula before integrating.
- **Volumes of revolution:** it is y^2 , not y — remember to square $f(x)$.
- **Differential equations:** after integrating, always apply the initial condition to fix C .
- **Check by differentiation** whenever time permits — a 10-second sanity check.

Typical OCR examiner complaints (from mark reports).

- Forgetting $+C$ in indefinite integrals — a reliable way to lose an easy mark.
- Changing the variable via substitution but forgetting to change the limits, or evaluating u -limits on an x -expression.
- Choosing u and dv the wrong way round in integration by parts and then getting stuck in a harder integral.
- Integrating $\cos^2 x$ as $\cos^3 x/3$ (wrong — not a power of x).
- In volumes of revolution, integrating y instead of y^2 , or forgetting the factor of π .
- For "area between a curve and the x-axis", failing to check whether the curve crosses the axis — taking the integral alone when the answer required the geometric area.
- In partial fractions, cover-up arithmetic errors, or decomposing a non-proper rational function (where $\deg P \geq \deg Q$) without first doing polynomial division.
- Solving a differential equation but forgetting to apply the initial condition — leaving C unspecified when the question demanded the particular solution.

Formula snapshot (memorise).

- Power rule: $\int x^n dx = x^{n+1}/(n+1) + C$ (for $n \neq -1$). $\int 1/x dx = \ln|x| + C$.
- Trig: $\int \sin x dx = -\cos x + C$; $\int \cos x dx = \sin x + C$; $\int \sec^2 x dx = \tan x + C$.
- Exp: $\int e^x dx = e^x + C$; $\int e^{kx} dx = (1/k)e^{kx} + C$.
- Linear inside: $\int f(ax+b) dx = (1/a)F(ax+b) + C$.
- Definite: $\int_a^b f(x) dx = F(b) - F(a)$.
- Substitution: $u = g(x)$, $du = g'(x) dx$. Change limits for definite.
- Parts: $\int u dv = uv - \int v du$. Choose u by LIATE.
- Partial fractions: decompose, then each piece integrates to $\ln$ or a power.
- $\cos^2 x = \frac{1}{2}(1 + \cos 2x)$; $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$.
- Volume of revolution: $V = \pi \int_a^b y^2 dx$ (about x-axis); $V = \pi \int_c^d x^2 dy$ (about y-axis).
- Separable ODE: $\int dy/g(y) = \int f(x) dx + C$; apply initial condition.